GREEEN tallgrass prairie restoration: RLQ analyses of pool size (Lux Arbor)

MB

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Preparation

```
library(here)
library(tidyverse)
library(ade4)
```

Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    id_plot = "f",
    year = "i",
    seeding_time = col_factor(
      levels = c("unseeded", "fall", "spring"), ordered = FALSE
      ),
    seeded_pool = col_factor(
```

```r
      levels = c("0", "6", "12", "18", "33")#, ordered = TRUE
      ),
    treatment_id = "f",
    richness_type = "f"
  )
) %>%
  filter(
    site == "Lux Arbor",
    richness_type == "seeded_richness",
    !(treatment_id %in% c("2", "4"))
  ) %>%
    mutate(
    treatment = str_c(seeding_time, seeded_pool, sep = "_"),
    treatment = factor(treatment)
  ) %>%
  select(
    id_plot_year, year, treatment, water_cap, cover_non_seeded,
    cover_seeded_grass
    )

traits <- read_csv(
  here("data", "processed", "data_processed_traits.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(seeded == "1", family != "Poaceae") %>%
  select(accepted_name, sla, height, seedmass) %>%
  mutate(seedmass = log(seedmass)) %>%
  drop_na()

species <- read_csv(
  here("data", "processed", "data_processed_species.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(plot_size %in% c("1", "25")) %>%
  mutate(abundance = 1) %>%
  select(id_plot_year, accepted_name, abundance) %>%
  semi_join(sites, by = "id_plot_year") %>%
  semi_join(traits, by = "accepted_name") %>%
  arrange(id_plot_year)
```

Load data

Checks

```r
sites %>%
  group_by(year) %>%
  count()
```

Plots per year

```
## # A tibble: 6 x 2
## # Groups:   year [6]
##    year     n
##   <int> <int>
```

```
## 1  2015    47
## 2  2016    48
## 3  2017    48
## 4  2018    48
## 5  2019    48
## 6  2025    48
```

```
species %>%
  group_by(id_plot_year) %>%
  count(name = "seeded_forb_richness") %>%
  ungroup() %>%
  count(seeded_forb_richness, name = "number_of_plots") %>%
  mutate(
    "plots_>=_1_species" = sum(number_of_plots),
    total_number_plots = 288
    )
```

Seeded species richness

```
## # A tibble: 16 x 4
##    seeded_forb_richness number_of_plots `plots_>=_1_species` total_number_plots
##                   <int>           <int>                <int>               <dbl>
##  1                    1              26                  215                 288
##  2                    2              22                  215                 288
##  3                    3              12                  215                 288
##  4                    4              16                  215                 288
##  5                    5               7                  215                 288
##  6                    6              19                  215                 288
##  7                    7              15                  215                 288
##  8                    8              32                  215                 288
##  9                    9              17                  215                 288
## 10                   10              16                  215                 288
## 11                   11              11                  215                 288
## 12                   12              13                  215                 288
## 13                   13               3                  215                 288
## 14                   14               3                  215                 288
## 15                   16               2                  215                 288
## 16                   17               1                  215                 288
```

```
traits %>% anti_join(species, by = "accepted_name")
```

Check non-occuring seeded species

```
## # A tibble: 6 x 4
##   accepted_name                   sla height seedmass
##   <chr>                         <dbl>  <dbl>    <dbl>
## 1 Eryngium yuccifolium           91.7  1.01     -5.57
## 2 Liatris cylindracea           110.   0.564    -5.73
## 3 Lupinus perennis              167.   0.326    -3.84
## 4 Silphium terebinthinaceum     171.   0.716    -4.53
## 5 Solidago juncea               130.   0.783    -7.79
## 6 Symphyotrichum oolentangiense 143.   0.362    -8.68
```

Finalize preparation

```
traits <- traits %>%
  semi_join(species, by = "accepted_name") %>%
  column_to_rownames(var = "accepted_name")

sites <- sites %>%
  semi_join(species, by = "id_plot_year") %>%
  column_to_rownames(var = "id_plot_year")

species <- species %>%
  pivot_wider(
    names_from = "accepted_name", values_from = "abundance", values_fill = 0
    ) %>%
  column_to_rownames(var = "id_plot_year")
```

Statistics

Descriptive analyses

```
sites %>%
  group_by(year) %>%
  count()
```

```
## # A tibble: 6 x 2
## # Groups:   year [6]
##    year     n
##   <int> <int>
## 1  2015    32
## 2  2016    36
## 3  2017    36
## 4  2018    34
## 5  2019    37
## 6  2025    40
```

Ordinations

```
dudi_l <- dudi.coa(species, scannf = FALSE)
dudi_r <- dudi.hillsmith(
  sites,
  row.w = dudi_l$lw,
  scannf = FALSE
  )
dudi_q <- dudi.pca(
  traits,
  row.w = dudi_l$cw,
  scannf = FALSE
  )
```

RLQ analyses

```r
m_rlq <- rlq(
  dudiR = dudi_r,
  dudiL = dudi_l,
  dudiQ = dudi_q,
  scannf = FALSE
  )
```

Global test

Model 2: species-environment relationship

Model 4: species-traits relationship

```r
randtest(m_rlq, nrepet = 999, modeltype = 6)
```

```
## class: krandtest lightkrandtest
## Monte-Carlo tests
## Call: randtest.rlq(xtest = m_rlq, nrepet = 999, modeltype = 6)
##
## Number of tests:   2
##
## Adjustment method for multiple comparisons:   none
## Permutation number:   999
##      Test        Obs   Std.Obs   Alter Pvalue
## 1 Model 2 0.06516472 15.495335 greater  0.001
## 2 Model 4 0.06516472 -1.423383 greater  0.944
```

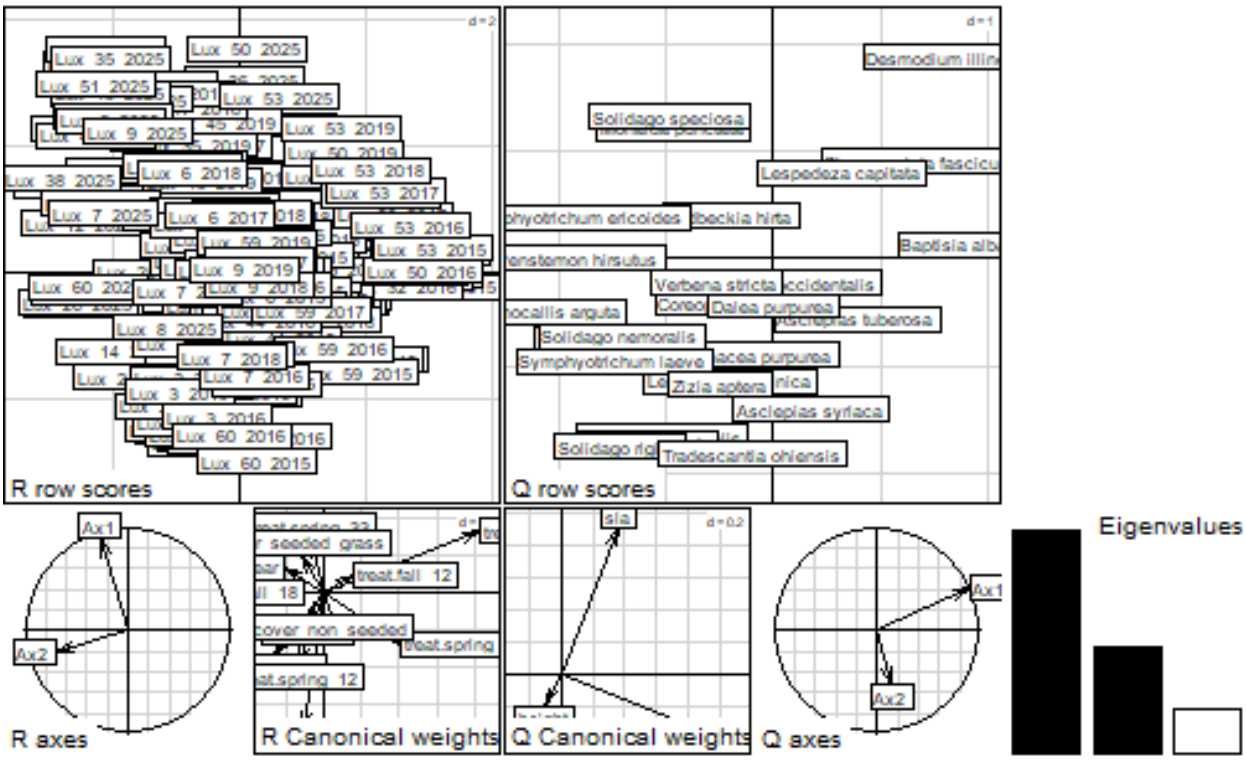
Traits by environment crossed table

```r
m_rlq$tab
```

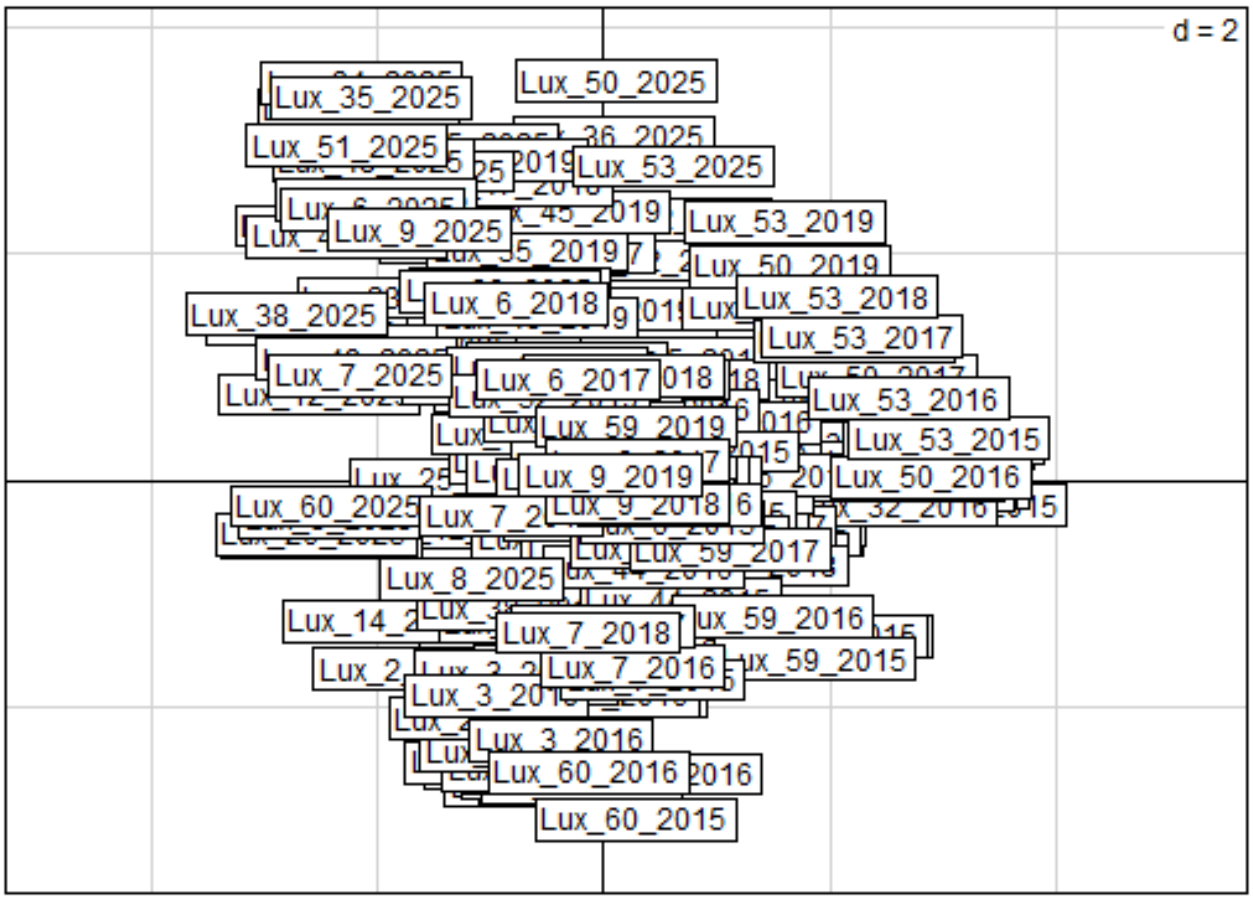
```
##                            sla       height    seedmass
## year                0.007631328  0.011771497 -0.11358152
## treat.fall_12       0.072025537  0.062790238  0.05886750
## treat.fall_18       0.004230655  0.100659720 -0.04665344
## treat.fall_33      -0.158346386 -0.007714225 -0.07982157
## treat.fall_6        0.234996250 -0.107810993  0.33955474
## treat.spring_12    -0.175102996 -0.100865799 -0.01703061
## treat.spring_18    -0.254376463  0.012612223  0.04138988
## treat.spring_33     0.060716117 -0.079957663 -0.06966565
## treat.spring_6     -0.088512527 -0.328282553  0.21596829
## water_cap          -0.063086842  0.005355516 -0.02363435
## cover_non_seeded   -0.027874648  0.021297546  0.03250372
## cover_seeded_grass  0.036315256 -0.026865687 -0.08247064
```

RLQ Plots

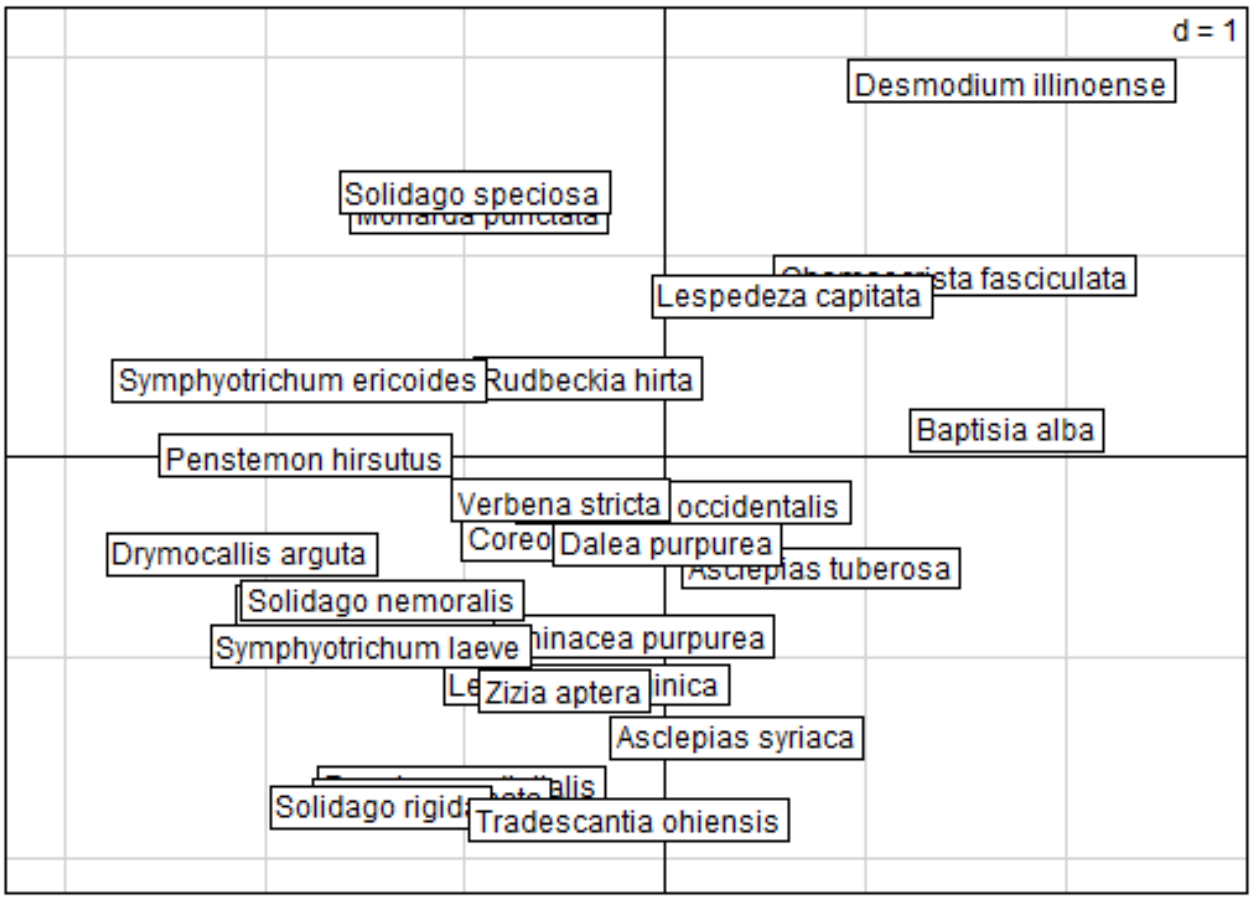
```r
plot(m_rlq)
```



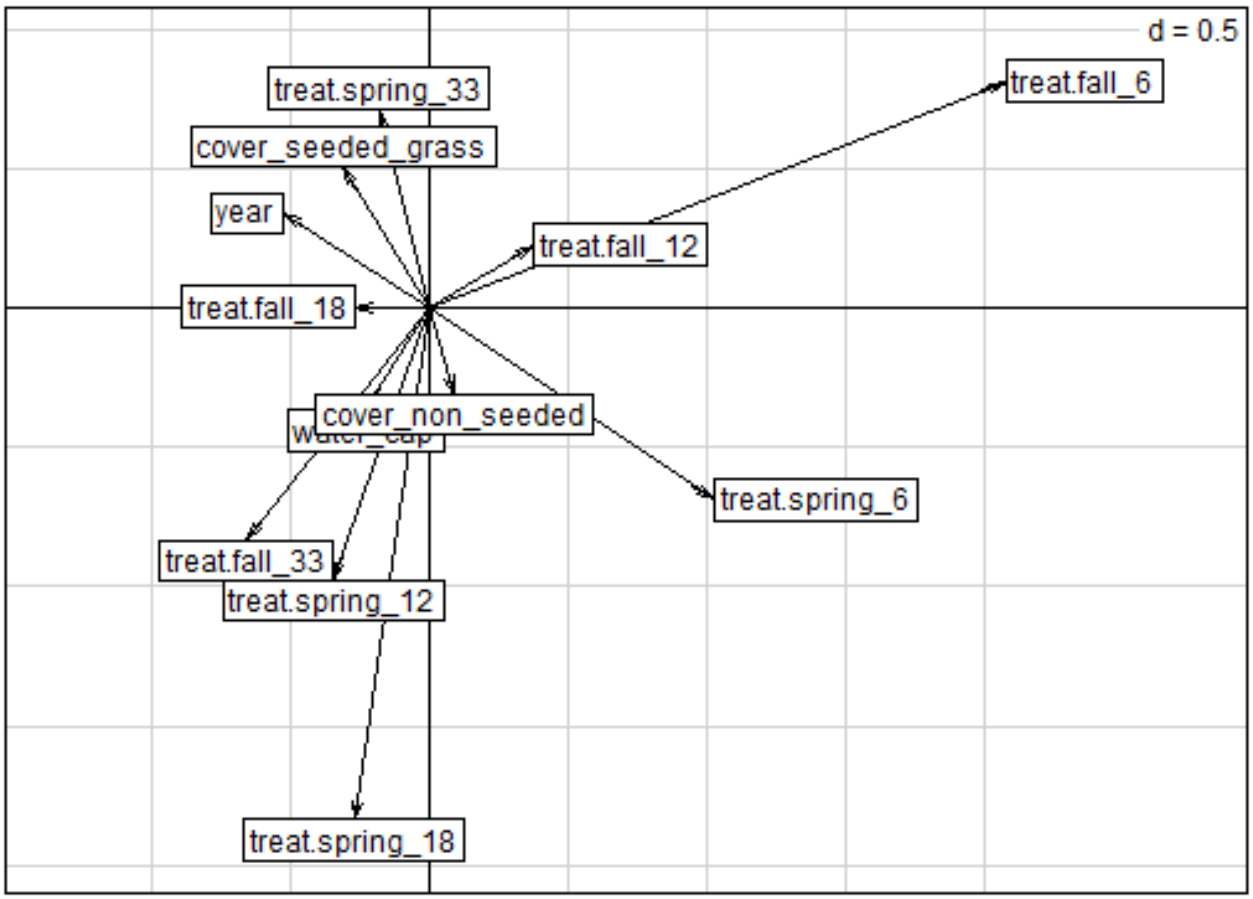

```
s.label(m_rlq$lR)
```



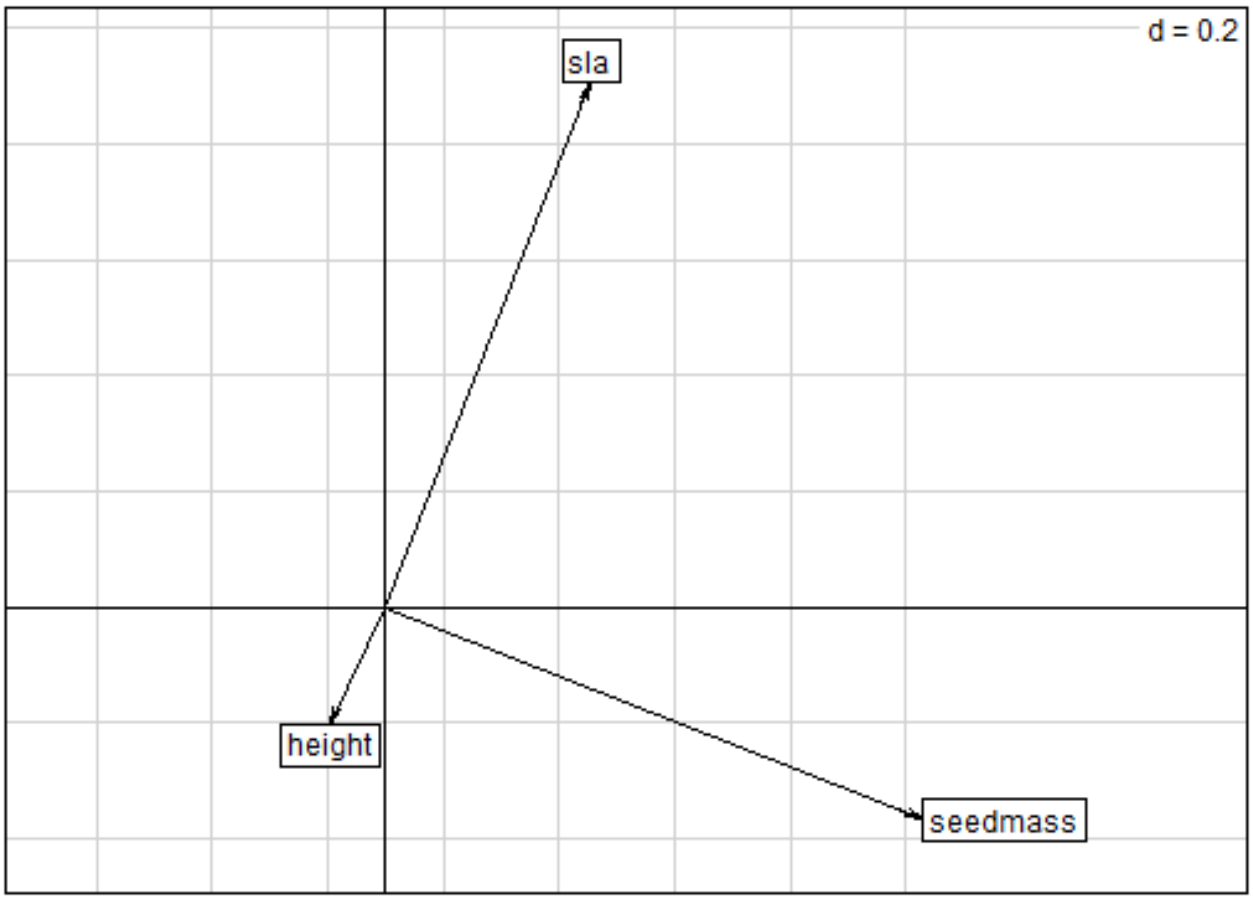

```
s.label(m_rlq$lQ)
```




```
s.arrow(m_rlq$l1)
```




```
s.arrow(m_rlq$c1)
```



Fourth-corner analysis

```
m_4th <- fourthcorner(
  tabR = sites,
  tabL = species,
  tabQ = traits,
  modeltype = 6,
  p.adjust.method.G = "none",
  p.adjust.method.D = "none",
  nrepet = 49999
)
m_4th # non-adjusted p-values
```

```
## Fourth-corner Statistics
## -------------------------
## Permutation method  Comb. 2 and 4  ( 49999  permutations)
##
## Adjustment method for multiple comparisons:   none
## call:  fourthcorner(tabR = sites, tabL = species, tabQ = traits, modeltype = 6,      nrepet = 49999,
##
## ---
##
##                              Test   Stat          Obs     Std.Obs     Alter
## 1                     year / sla      r  0.007631328  0.10147418 two-sided
## 2            treat.fall_12 / sla Homog.  0.136091209 -0.86488872      less
```

```
## 3            treat.fall_18 / sla Homog.  0.173482783  0.31822214      less
## 4            treat.fall_33 / sla Homog.  0.317678832  7.90488950      less
## 5             treat.fall_6 / sla Homog.  0.068520525 -0.21931789      less
## 6          treat.spring_12 / sla Homog.  0.012596403 -0.12085137      less
## 7          treat.spring_18 / sla Homog.  0.003226291 -0.55546624      less
## 8          treat.spring_33 / sla Homog.  0.266516085  5.60962271      less
## 9           treat.spring_6 / sla Homog.  0.007621895 -1.13224737      less
## 10               water_cap / sla      r -0.063086842 -1.58079536 two-sided
## 11        cover_non_seeded / sla      r -0.027874648 -0.57107201 two-sided
## 12      cover_seeded_grass / sla      r  0.036315256  0.58792971 two-sided
## 13                 year / height      r  0.011771497  0.16118651 two-sided
## 14        treat.fall_12 / height Homog.  0.167892652  0.02847528      less
## 15        treat.fall_18 / height Homog.  0.196255685  1.40538202      less
## 16        treat.fall_33 / height Homog.  0.342953252  8.81623084      less
## 17         treat.fall_6 / height Homog.  0.053912027 -0.91810482      less
## 18      treat.spring_12 / height Homog.  0.014153421  0.07248295      less
## 19      treat.spring_18 / height Homog.  0.007085043  0.69521915      less
## 20      treat.spring_33 / height Homog.  0.197020800  1.93752253      less
## 21       treat.spring_6 / height Homog.  0.013460441 -0.13911853      less
## 22            water_cap / height      r  0.005355516  0.12224971 two-sided
## 23     cover_non_seeded / height      r  0.021297546  0.41054650 two-sided
## 24   cover_seeded_grass / height      r -0.026865687 -0.43028789 two-sided
## 25               year / seedmass      r -0.113581516 -1.52833653 two-sided
## 26      treat.fall_12 / seedmass Homog.  0.248619537  4.18305543      less
## 27      treat.fall_18 / seedmass Homog.  0.249712572  4.22298841      less
## 28      treat.fall_33 / seedmass Homog.  0.242926729  4.18032188      less
## 29       treat.fall_6 / seedmass Homog.  0.028121329 -1.84955806      less
## 30    treat.spring_12 / seedmass Homog.  0.002718566 -1.51222606      less
## 31    treat.spring_18 / seedmass Homog.  0.001580264 -1.30708872      less
## 32    treat.spring_33 / seedmass Homog.  0.205259180  2.48297974      less
## 33     treat.spring_6 / seedmass Homog.  0.006223584 -1.74011133      less
## 34          water_cap / seedmass      r -0.023634347 -0.57821351 two-sided
## 35   cover_non_seeded / seedmass      r  0.032503725  0.63976744 two-sided
## 36 cover_seeded_grass / seedmass      r -0.082470643 -1.32764371 two-sided
##     Pvalue
## 1  0.92642
## 2  0.19902
## 3   0.6286
## 4        1
## 5  0.46358
## 6  0.51566
## 7  0.32686
## 8        1
## 9  0.13942
## 10 0.11334
## 11 0.59472
## 12  0.5766
## 13 0.87964
## 14  0.5152
## 15 0.91866
## 16       1
## 17  0.2004
## 18   0.575
## 19 0.76694
```

```
## 20  0.9713
## 21 0.46388
## 22 0.90396
## 23  0.6959
## 24 0.68152
## 25 0.13028
## 26       1
## 27       1
## 28 0.99998
## 29 0.00862 **
## 30 0.02086  *
## 31 0.06296  .
## 32 0.99388
## 33 0.02764  *
## 34 0.56762
## 35  0.5414
## 36 0.19306
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
m_4th_adj <- p.adjust.4thcorner(
  m_4th,
  p.adjust.method.G = "fdr",
  p.adjust.method.D = "fdr",
  p.adjust.D = "global"
)
m_4th_adj
```

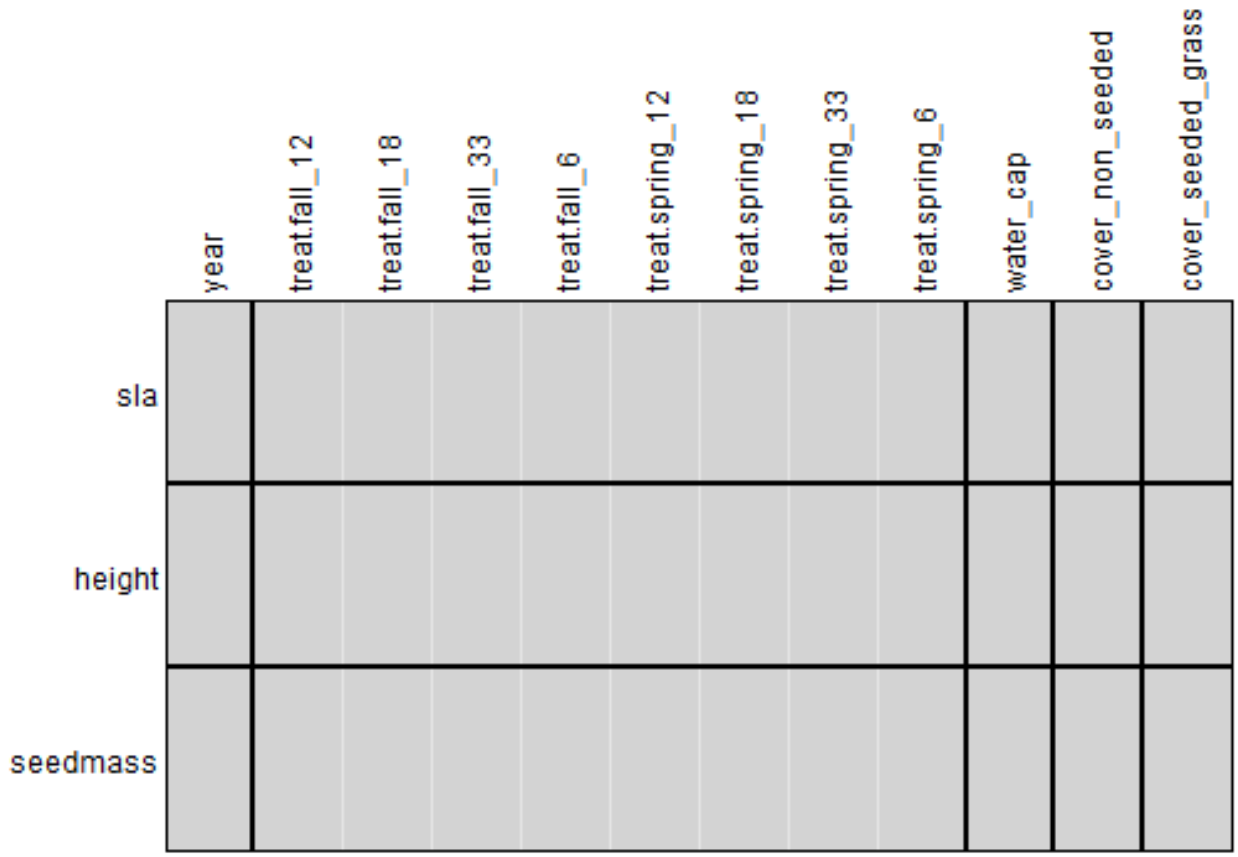
```
## Fourth-corner Statistics
## ------------------------
## Permutation method  Comb. 2 and 4  ( 49999  permutations)
##
## Adjustment method for multiple comparisons:   fdr
## call:  fourthcorner(tabR = sites, tabL = species, tabQ = traits, modeltype = 6,      nrepet = 49999,
##
## ---
##
##                              Test   Stat          Obs     Std.Obs     Alter
## 1                     year / sla      r  0.007631328  0.10147418 two-sided
## 2            treat.fall_12 / sla Homog.  0.136091209 -0.86488872      less
## 3            treat.fall_18 / sla Homog.  0.173482783  0.31822214      less
## 4            treat.fall_33 / sla Homog.  0.317678832  7.90488950      less
## 5             treat.fall_6 / sla Homog.  0.068520525 -0.21931789      less
## 6          treat.spring_12 / sla Homog.  0.012596403 -0.12085137      less
## 7          treat.spring_18 / sla Homog.  0.003226291 -0.55546624      less
## 8          treat.spring_33 / sla Homog.  0.266516085  5.60962271      less
## 9           treat.spring_6 / sla Homog.  0.007621895 -1.13224737      less
## 10               water_cap / sla      r -0.063086842 -1.58079536 two-sided
## 11        cover_non_seeded / sla      r -0.027874648 -0.57107201 two-sided
## 12      cover_seeded_grass / sla      r  0.036315256  0.58792971 two-sided
## 13                 year / height      r  0.011771497  0.16118651 two-sided
## 14        treat.fall_12 / height Homog.  0.167892652  0.02847528      less
## 15        treat.fall_18 / height Homog.  0.196255685  1.40538202      less
## 16        treat.fall_33 / height Homog.  0.342953252  8.81623084      less
```

```
## 17         treat.fall_6 / height Homog.  0.053912027 -0.91810482      less
## 18      treat.spring_12 / height Homog.  0.014153421  0.07248295      less
## 19      treat.spring_18 / height Homog.  0.007085043  0.69521915      less
## 20      treat.spring_33 / height Homog.  0.197020800  1.93752253      less
## 21       treat.spring_6 / height Homog.  0.013460441 -0.13911853      less
## 22            water_cap / height      r  0.005355516  0.12224971 two-sided
## 23     cover_non_seeded / height      r  0.021297546  0.41054650 two-sided
## 24   cover_seeded_grass / height      r -0.026865687 -0.43028789 two-sided
## 25               year / seedmass      r -0.113581516 -1.52833653 two-sided
## 26      treat.fall_12 / seedmass Homog.  0.248619537  4.18305543      less
## 27      treat.fall_18 / seedmass Homog.  0.249712572  4.22298841      less
## 28      treat.fall_33 / seedmass Homog.  0.242926729  4.18032188      less
## 29       treat.fall_6 / seedmass Homog.  0.028121329 -1.84955806      less
## 30    treat.spring_12 / seedmass Homog.  0.002718566 -1.51222606      less
## 31    treat.spring_18 / seedmass Homog.  0.001580264 -1.30708872      less
## 32    treat.spring_33 / seedmass Homog.  0.205259180  2.48297974      less
## 33     treat.spring_6 / seedmass Homog.  0.006223584 -1.74011133      less
## 34          water_cap / seedmass      r -0.023634347 -0.57821351 two-sided
## 35   cover_non_seeded / seedmass      r  0.032503725  0.63976744 two-sided
## 36 cover_seeded_grass / seedmass      r -0.082470643 -1.32764371 two-sided
##     Pvalue        Pvalue.adj
## 1  0.92642                 1
## 2  0.19902           0.72144
## 3   0.6286                 1
## 4        1                 1
## 5  0.46358                 1
## 6  0.51566                 1
## 7  0.32686                 1
## 8        1                 1
## 9  0.13942 0.717017142857143
## 10 0.11334 0.717017142857143
## 11 0.59472                 1
## 12  0.5766                 1
## 13 0.87964                 1
## 14  0.5152                 1
## 15 0.91866                 1
## 16       1                 1
## 17  0.2004           0.72144
## 18   0.575                 1
## 19 0.76694                 1
## 20  0.9713                 1
## 21 0.46388                 1
## 22 0.90396                 1
## 23  0.6959                 1
## 24 0.68152                 1
## 25 0.13028 0.717017142857143
## 26       1                 1
## 27       1                 1
## 28 0.99998                 1
## 29 0.00862           0.31032
## 30 0.02086           0.33168
## 31 0.06296           0.56664
## 32 0.99388                 1
## 33 0.02764           0.33168
```

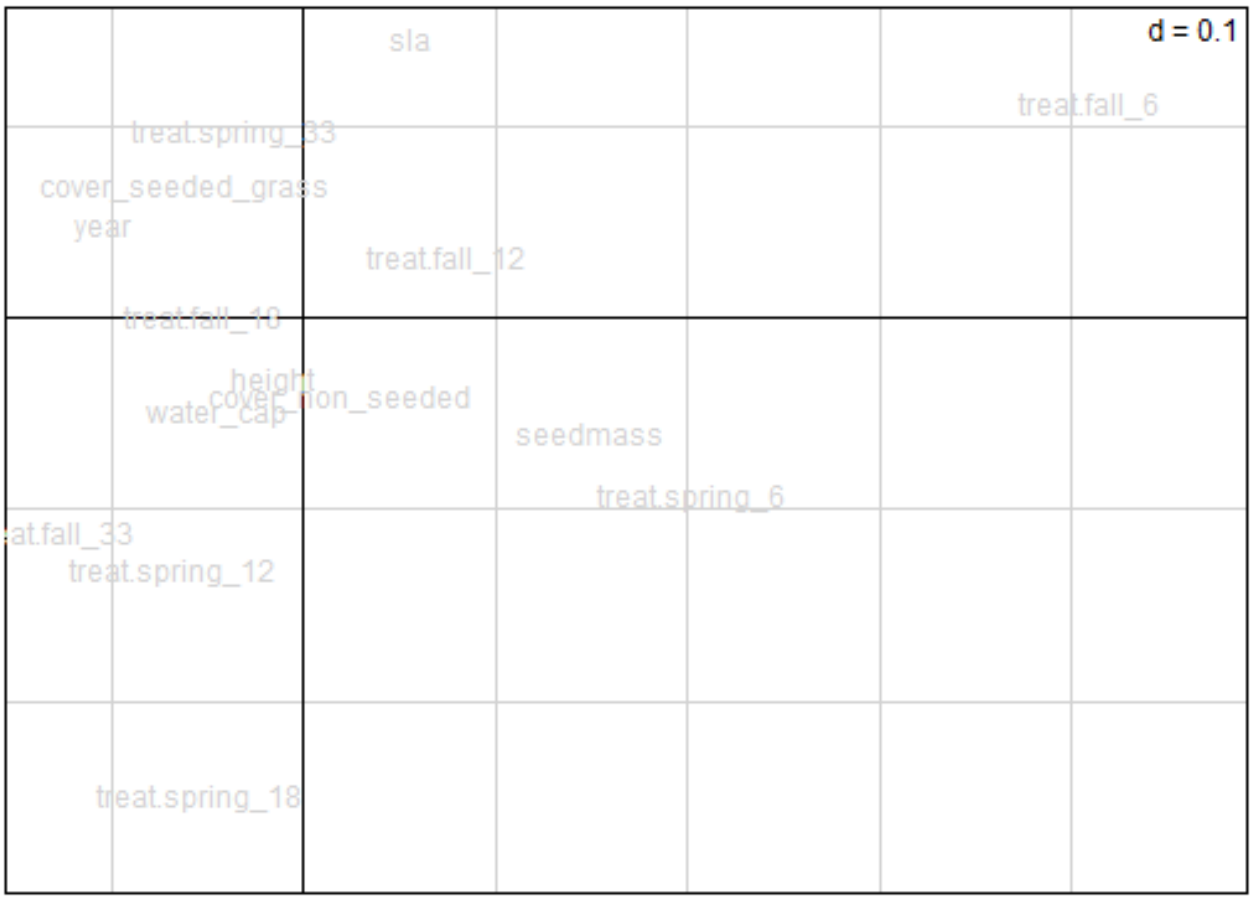
```
## 34 0.56762                 1
## 35  0.5414                 1
## 36 0.19306           0.72144
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Plot

```r
plot(m_4th_adj, alpha = 0.05, stat = "D2")
```




```r
plot(m_4th_adj,
  x.rlq = m_rlq,
  alpha = 0.05,
  stat = "D2",
  type = "biplot"
)
```



Save

Session info

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] ade4_1.7-23     lubridate_1.9.4 forcats_1.0.0   stringr_1.5.1
```

```
##  [5] dplyr_1.1.4     purrr_1.1.0     readr_2.1.5     tidyr_1.3.1
##  [9] tibble_3.3.0    ggplot2_3.5.2   tidyverse_2.0.0 here_1.0.1
## [13] tinytex_0.57
##
## loaded via a namespace (and not attached):
##  [1] bit_4.6.0          gtable_0.3.6       crayon_1.5.3       compiler_4.5.1
##  [5] Rcpp_1.1.0         tidyselect_1.2.1   parallel_4.5.1     scales_1.4.0
##  [9] yaml_2.3.10        fastmap_1.2.0      R6_2.6.1           generics_0.1.4
## [13] knitr_1.50         MASS_7.3-65        rprojroot_2.1.1    pillar_1.11.0
## [17] RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.1.6        utf8_1.2.6
## [21] stringi_1.8.7      xfun_0.52          bit64_4.6.0-1      timechange_0.3.0
## [25] cli_3.6.5          withr_3.0.2        magrittr_2.0.3     digest_0.6.37
## [29] grid_4.5.1         vroom_1.6.5        rstudioapi_0.17.1  hms_1.1.3
## [33] lifecycle_1.0.4    vctrs_0.6.5        evaluate_1.0.4     glue_1.8.0
## [37] farver_2.1.2       rmarkdown_2.29     tools_4.5.1        pkgconfig_2.0.3
## [41] htmltools_0.5.8.1
```